

Student Name: Keivan Jamali
Student ID: 403209743
Instructor: Dr. Shafahi
Date: 1405/03/06



Sharif University
of Technology

Airport (20_582)

Homework 5

Problem 1

There is an airplane with the following properties:

- V : airspeed (mile/hr)
- v : airspeed (ft/sec)
- $\rho_0 = 0.0024$ slugs/ft³ : air density at sea level
- $\rho_{36000} = 0.00072$ slugs/ft³ : air density at 36,000 ft
- $c = 0.8$ lbs/(lb · hr) : specific fuel consumption (SFC)
- w_1 : final weight
- w_0 : initial weight
- R : range (mile) = $\frac{V}{c} \left(\frac{L}{D} \right) \ln \left(\frac{w_0}{w_1} \right) = 2.3 \frac{V}{c} \left(\frac{L}{D} \right) \log_{10} \left(\frac{w_0}{w_1} \right)$
- $S = 3000$ ft² : wing planform area
- $S_t = 900$ ft² : tail planform area
- $b = 150$ ft : wingspan
- $l = 150$ ft : aircraft length
- $d_f = 12$ ft : fuselage diameter
- $d_e = 5$ ft : engine diameter
- h : altitude in ft
- C'' : fuel consumption rate (lb/mile)
- a : acceleration
- G : glide distance
- $f = 0.02$: rolling friction coefficient
- f_b : braking friction coefficient

- $w_g = 330000$ lbs : gross takeoff weight (GTOW)
- $w_e = 150000$ lbs : operating empty weight (OEW)
- $w_f = 150000$ lbs : maximum fuel weight
- $w_r = 15000$ lbs : reserve fuel weight
- $w_p = 100000$ lbs : payload weight ($w_g - w_e - w_f$)
- $n = 4$: number of engines
- $T_0 = 4 \times 15000$ lbs : total thrust at sea level
- $T_{36000} = 4 \times 4000$ lbs : total thrust at 36,000 ft altitude
- $C_{L_{max}} = 1.5$: maximum lift coefficient (for landing and takeoff)
- $C_D = C_{DP} + C_{DI}$: total drag coefficient
- $C_{DP_{tail}} = 0.01$: parasite drag coefficient of the tail
- $C_{DP_{wing}} = 0.0055$: parasite drag coefficient of the wing
- $C_{DI} = \frac{C_L^2}{\pi AR}$: induced drag coefficient
- $AR = \frac{b^2}{S}$: aspect ratio
- $L = C_L \cdot \left(\frac{\rho}{2}\right) \cdot V^2 \cdot S$: Lift force
- $D = C_D \cdot \left(\frac{\rho}{2}\right) \cdot V^2 \cdot S$: Drag force

We want this airplane to move the maximum possible baggage per time from point A to point D. The airplane may stop at B or C for refueling. Each stop (at A, B, C, or D) causes a 1.5-hour delay. Assume the cruise speed at 36,000 ft is 500 miles per hour.

The available routes and distances (in miles) are shown in Figure 1.

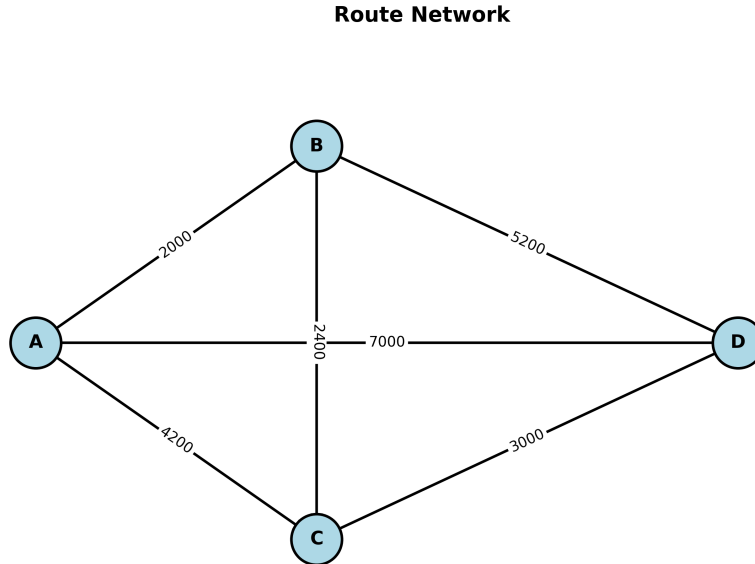


Figure 1: Route network with distances in miles.

Find:

1. Best path (for maximum baggage per day)
2. Maximum output (pounds per day)
3. Fuel consumption (pounds of fuel per pound of baggage delivered to D)
4. For the purpose of better efficiency, redesign the path to minimize fuel consumption (pounds of fuel per ton of baggage delivered to D)

Solution of Problem 1

0.0.1 Step 1: Aerodynamic Characteristics

The aspect ratio of the wing is:

$$AR = \frac{b^2}{S}$$

Substituting the given values:

$$AR = \frac{150^2}{3000} = 7.5$$

The parasite drag coefficient is:

$$C_{DP} = C_{DP_{tail}} + C_{DP_{wing}}$$

$$C_{DP} = 0.01 + 0.0055 = 0.0155$$

The induced drag coefficient is:

$$C_{DI} = \frac{C_L^2}{\pi AR}$$

The total drag coefficient becomes:

$$C_D = C_{DP} + C_{DI}$$

For maximum range, the airplane must operate at maximum aerodynamic efficiency:

$$\left(\frac{L}{D}\right)_{\max} = \frac{1}{2\sqrt{C_{DP}/(\pi AR)}}$$

Substituting the values:

$$\left(\frac{L}{D}\right)_{\max} = \frac{1}{2\sqrt{0.0155/(\pi \times 7.5)}}$$

$$\boxed{\left(\frac{L}{D}\right)_{\max} \approx 19.5}$$

0.0.2 Step 2: Breguet Range Equation

The aircraft range is given by the Breguet equation:

$$R = \frac{V}{c} \left(\frac{L}{D} \right) \ln \left(\frac{w_0}{w_1} \right)$$

where:

$$V = 500 \text{ mile/hr}$$

$$c = 0.8$$

$$\frac{L}{D} = 19.5$$

Define:

$$K = \frac{V}{c} \left(\frac{L}{D} \right)$$

Thus:

$$K = \frac{500}{0.8} \times 19.5$$

$$\boxed{K \approx 12187.5}$$

The maximum takeoff weight is:

$$w_0 = 330000 \text{ lb}$$

The final landing weight is:

$$w_1 = w_e + w_r + P$$

where P is the payload.

Since:

$$w_e = 150000 \text{ lb}$$

$$w_r = 15000 \text{ lb}$$

then:

$$w_1 = 165000 + P$$

Substituting into the Breguet equation:

$$R = K \ln \left(\frac{330000}{165000 + P} \right)$$

Solving for payload:

$$\boxed{P = \frac{330000}{e^{R/K}} - 165000}$$

This equation gives the maximum payload that can be transported for a given flight distance.

0.0.3 Step 3: Maximum Payload for Each Route

Using the previous equation:

$$P = \frac{330000}{e^{R/12187.5}} - 165000$$

the maximum payload for each route is calculated.

Route	Distance (mile)	Maximum Payload (lb)
AB	2000	115043
BC	2400	105998
CD	3000	92976
AC	4200	68779
BD	5200	50358
AD	7000	20781

0.0.4 Step 4: Evaluate Possible Paths

We now calculate the payload throughput for each path.

Path 1: $A \rightarrow B \rightarrow C \rightarrow D$ The payload is limited by the smallest payload among all legs:

$$P_{max} = \min(115043, 105998, 92976)$$

$$P_{max} = 92976 \text{ lb}$$

Total flight distance:

$$R_{total} = 2000 + 2400 + 3000$$

$$R_{total} = 7400 \text{ mile}$$

Flight time:

$$t_{flight} = \frac{7400}{500} = 14.8 \text{ hr}$$

There are four airport operations:

$$A, B, C, D$$

Each operation causes a 1.5 hr delay:

$$t_{ground} = 4 \times 1.5 = 6 \text{ hr}$$

Total transportation time:

$$t_{total} = 14.8 + 6$$

$$t_{total} = 20.8 \text{ hr}$$

Trips per day:

$$N = \frac{24}{20.8} = 1.154$$

Payload transported per day:

$$Q = 92976 \times 1.154$$

$$Q \approx 107000 \text{ lb/day}$$

Path 2: $A \rightarrow C \rightarrow D$ Payload limitation:

$$P_{max} = \min(68779, 92976)$$

$$P_{max} = 68779 \text{ lb}$$

Total distance:

$$4200 + 3000 = 7200 \text{ mile}$$

Flight time:

$$\frac{7200}{500} = 14.4 \text{ hr}$$

Ground delay:

$$3 \times 1.5 = 4.5 \text{ hr}$$

Total time:

$$18.9 \text{ hr}$$

Payload/day:

$$68779 \times \frac{24}{18.9}$$

$$Q \approx 87300 \text{ lb/day}$$

Path 3: $A \rightarrow B \rightarrow D$ Payload limitation:

$$P_{max} = \min(115043, 50358)$$

$$P_{max} = 50358 \text{ lb}$$

Total distance:

$$2000 + 5200 = 7200 \text{ mile}$$

Flight time:

$$14.4 \text{ hr}$$

Ground delay:

$$4.5 \text{ hr}$$

Total time:

$$18.9 \text{ hr}$$

Payload/day:

$$50358 \times \frac{24}{18.9}$$

$$Q \approx 63900 \text{ lb/day}$$

Path 4: $A \rightarrow D$ Maximum payload:

$$P_{max} = 20781 \text{ lb}$$

Flight time:

$$\frac{7000}{500} = 14 \text{ hr}$$

Ground delay:

$$2 \times 1.5 = 3 \text{ hr}$$

Total time:

$$17 \text{ hr}$$

Payload/day:

$$20781 \times \frac{24}{17}$$

$$Q \approx 29300 \text{ lb/day}$$

Step 5: Best Path Comparing all paths:

Path	Payload per Day (lb/day)
$A \rightarrow B \rightarrow C \rightarrow D$	107000
$A \rightarrow C \rightarrow D$	87300
$A \rightarrow B \rightarrow D$	63900
$A \rightarrow D$	29300

Therefore, the best path is:

$$A \rightarrow B \rightarrow C \rightarrow D$$

0.0.5 Step 6: Fuel Consumption

The fuel consumption for one trip is:

$$w_f = w_0 - w_1$$

For the optimal route:

$$P = 92976 \text{ lb}$$

Thus:

$$w_1 = 165000 + 92976$$

$$w_1 = 257976 \text{ lb}$$

The initial weight is:

$$w_0 = 330000 \text{ lb}$$

Therefore:

$$w_f = 330000 - 257976$$

$$w_f = 72024 \text{ lb}$$

Fuel consumption per payload pound:

$$\frac{72024}{92976}$$

$$0.775 \text{ lb fuel/lb payload}$$

Fuel consumption per ton of payload:

$$0.775 \times 2000$$

$$1550 \text{ lb fuel/ton payload}$$

0.1 Final Answers

1. Best path:

$$A \rightarrow B \rightarrow C \rightarrow D$$

2. Maximum payload transportation rate:

$$1.07 \times 10^5 \text{ lb/day}$$

3. Fuel consumption:

$$0.775 \text{ lb fuel/lb payload}$$

4. The most fuel-efficient route is also:

$$A \rightarrow B \rightarrow C \rightarrow D$$

Problem 2

Determine the minimum runway (or pavement) width required for a Boeing 757-200 aircraft to perform a 180° turn (U-turn).

The following aircraft data are given:

- Aircraft length: 47.3 m
- Wingspan: 38.1 m
- Wheelbase (distance between nose gear and main landing gear): 18.29 m
- Maximum nose wheel steering angle (assumed): 65°

Answer

To estimate the required pavement width for a U-turn, the turning radius of the aircraft must first be calculated.

Using the vehicle turning geometry relation:

$$R = \frac{L}{\tan \theta}$$

where:

- R = turning radius
- $L = 18.29$ m = wheelbase
- $\theta = 65^\circ$ = maximum steering angle

First, calculate:

$$\tan(65^\circ) \approx 2.145$$

Therefore:

$$R = \frac{18.29}{2.145}$$

$$R \approx 8.52 \text{ m}$$

This radius corresponds approximately to the landing gear turning path. However, the aircraft fuselage and wings extend beyond this path, therefore additional swept width must be considered.

For large commercial aircraft, the required pavement width can be approximated as:

$$W \approx 2R + B$$

where:

- W = required pavement width
- $R = 8.52$ m
- B = additional swept width caused by fuselage and wing overhang

For the Boeing 757-200, the additional swept width is approximately:

$$B \approx 19.5 \text{ m}$$

Thus:

$$W \approx 2(8.52) + 19.5$$

$$W \approx 17.04 + 19.5$$

$$W \approx 36.5 \text{ m}$$

Therefore, the minimum pavement width required for a Boeing 757-200 to perform a 180° turn is approximately:

$$W \approx 36.6 \text{ m}$$

which is equivalent to:

$$120 \text{ ft}$$

It should be noted that this value is also available in Boeing airport planning references and aviation engineering sources, where the published minimum pavement width for a Boeing 757-200 180° turn is approximately 36.6 m (120 ft).

Problem 3

The table below presents the profile specifications of a proposed airport runway site. A runway is required for this airport with a **basic length** of 8000 ft. Apply the following corrections to the runway length:

- Add 10% to the runway length for each 1% of effective gradient.
- Add 7% to the runway length for every 1000 ft elevation above mean sea level.

You are required to design an airport runway at this location in such a way that the FAA regulations are satisfied while minimizing the overall cost. The following cost data are provided:

- Pavement cost: \$20 per square foot
- Cutting or filling cost: \$0.03 per cubic foot
- Borrowing fill material or hauling excess cut material outside the site: \$0.05 per cubic foot

Show the ground profile, all calculations, and explain how your design satisfies FAA regulations. (Minor approximations are acceptable.)

Site Profile Data

Station	Elevation (ft)
0	1020
20	980
40	980
60	1000
80	1040
100	1020
120	990
140	1000
160	1040

Solution

The required runway length must first be corrected for both airport elevation and effective gradient according to FAA recommendations.

1. Elevation Correction

The average elevation of the site is calculated from the given profile data:

$$\bar{E} = \frac{1020 + 980 + 980 + 1000 + 1040 + 1020 + 990 + 1000 + 1040}{9}$$

$$\bar{E} = \frac{9070}{9} = 1007.8 \text{ ft}$$

Therefore, the average airport elevation is approximately:

$$\bar{E} \approx 1008 \text{ ft}$$

FAA requires a 7% increase in runway length for every 1000 ft elevation above mean sea level:

$$\text{Elevation correction} = 7\% \times \frac{1008}{1000} = 7.056\%$$

The corrected runway length becomes:

$$L_1 = 8000(1 + 0.07056)$$

$$L_1 = 8564.5 \text{ ft}$$

2. Effective Gradient Correction

The effective gradient is defined as:

$$G = \frac{\text{Maximum elevation difference}}{\text{Runway length}} \times 100$$

From the profile data:

$$E_{\max} = 1040 \text{ ft}$$

$$E_{\min} = 980 \text{ ft}$$

Thus,

$$\Delta h = 1040 - 980 = 60 \text{ ft}$$

Using the corrected runway length:

$$G = \frac{60}{8564.5} \times 100$$

$$G \approx 0.70\%$$

FAA requires a 10% increase in runway length for each 1% effective gradient. Therefore:

$$\text{Gradient correction} = 0.70 \times 10\% = 7.0\%$$

The final corrected runway length becomes:

$$L_f = 8564.5(1 + 0.07)$$

$$L_f = 9163.9 \text{ ft}$$

Therefore, the required runway length is approximately:

$$\boxed{9200 \text{ ft}}$$

3. Proposed Runway Profile

The existing ground profile contains relatively steep changes in elevation, which may violate FAA longitudinal slope recommendations. Therefore, grading operations are required to create a smoother runway profile.

A proposed economical runway profile is shown below:

This profile smooths the runway grades and reduces abrupt elevation changes, helping satisfy FAA slope limitations.

Station	Existing Elevation (ft)	Proposed Elevation (ft)	Cut / Fill (ft)
0	1020	1005	Cut 15
20	980	1000	Fill 20
40	980	1000	Fill 20
60	1000	1005	Fill 5
80	1040	1010	Cut 30
100	1020	1005	Cut 15
120	990	1000	Fill 10
140	1000	1005	Fill 5
160	1040	1010	Cut 30

4. Earthwork Evaluation

Approximate total cut:

$$15 + 30 + 15 + 30 = 90$$

Approximate total fill:

$$20 + 20 + 5 + 10 + 5 = 60$$

Thus, there is approximately:

$$90 - 60 = 30$$

units of excess cut material that must be hauled outside the site.

Because the proposed design keeps the cut and fill quantities relatively close, hauling and borrowing costs are minimized.

5. Pavement Cost

Assuming a standard runway width of:

$$W = 150 \text{ ft}$$

Runway pavement area:

$$A = 9200 \times 150$$

$$A = 1,380,000 \text{ ft}^2$$

Given the pavement cost of \$20 per square foot:

$$\text{Pavement Cost} = 1,380,000 \times 20$$

$$= \$27,600,000$$

Problem 4

A sketch of the obstruction surfaces defined in FAR Part 77 is shown in Fig. B-1. The obstruction surfaces are to be specified for approaches to a 10,000-ft runway designed for use by transport aircraft for precision-instrument operations. The elevation of the runway is 642.5 ft AMSL.

- (a) Determine the coordinates of points A, B, C, D, E, F, and G shown in Fig. B-1.
- (b) Determine whether an object which is 280 ft high located at a point 12,000 ft from the midpoint of the runway along the extended runway centerline penetrates the FAR Part 77 obstruction surfaces.
- (c) Determine the maximum elevation of a tower located at a point 2000 ft from the midpoint of the runway measured along the runway centerline and offset a distance of 8000 ft from the runway centerline, if it is not to penetrate the FAR Part 77 obstruction surfaces.

Answer

Assume the origin is located at the midpoint of the runway:

$$(0, 0, 642.5)$$

where the x -axis is along the runway centerline and the y -axis is perpendicular to the runway.

For a precision-instrument runway according to FAR Part 77:

- Horizontal surface elevation:

$$642.5 + 150 = 792.5 \text{ ft}$$

- Conical surface slope:

$$20 : 1$$

- Approach surface:

- Inner width = 1000 ft
- Outer width = 16000 ft
- Length = 50000 ft
- Slope = 50 : 1 for first 10000 ft, then 40 : 1

(a) Coordinates of the points

$$A = (0, 5000, 792.5)$$

$$B = (0, 9000, 992.5)$$

For point C, the approach surface reaches the horizontal surface elevation after:

$$150 \times 50 = 7500 \text{ ft}$$

from the start of the approach surface:

$$x_C = 5200 + 7500 = 12700$$

The half-width at this point is:

$$500 + \frac{7500}{50000} \times 7500 = 1625$$

Hence:

$$C = (12700, 1625, 792.5)$$

For point G :

$$x_G = 5200 + 1000 = 6200$$

$$z_G = 642.5 + \frac{1000}{50} = 662.5$$

$$y_G = 500 + \frac{7500}{50000} \times 1000 = 650$$

Thus:

$$G = (6200, 650, 662.5)$$

For point D :

$$x_D = 5200 + 50000 = 55200$$

Elevation rise:

$$\frac{10000}{50} + \frac{40000}{40} = 200 + 1000 = 1200$$

Therefore:

$$D = (55200, 0, 1842.5)$$

Point E :

$$E = (55200, 8000, 1842.5)$$

For point F , the transitional surface rises at 7 : 1:

$$z_F = 1842.5 + \frac{5000}{7} = 2556.8$$

Hence:

$$F = (55200, 13000, 2556.8)$$

(b) Obstruction check The object is located 12000 ft from the runway midpoint:

$$x = 12000$$

Distance from beginning of approach surface:

$$12000 - 5200 = 6800 \text{ ft}$$

Allowed elevation rise:

$$\frac{6800}{50} = 136 \text{ ft}$$

Maximum permitted elevation:

$$642.5 + 136 = 778.5 \text{ ft}$$

Object elevation:

$$642.5 + 280 = 922.5 \text{ ft}$$

Since:

$$922.5 > 778.5$$

the object penetrates the FAR Part 77 obstruction surface by:

$$922.5 - 778.5 = 144 \text{ ft}$$

(c) **Maximum tower elevation** Tower location:

$$(2000, 8000)$$

Distance from runway center:

$$r = \sqrt{2000^2 + 8000^2} = 8246 \text{ ft}$$

Since:

$$8246 < 10000$$

the point lies inside the horizontal surface.

Maximum allowable elevation:

$$792.5 \text{ ft AMSL}$$

Maximum tower height:

$$792.5 - 642.5 = 150 \text{ ft}$$

Therefore, the maximum allowable tower height is:

$$150 \text{ ft}$$

Problem 5

A rough sizing of the airport site is to be made for an airport with parallel runways to be operated in an IFR environment with arrivals on one runway and departures on the other runway. The thresholds of the runways are not staggered. The airside system is to consist of parallel taxiways serving each runway, the terminal complex on one side of the parallel and the property boundary on the other side. The design aircraft is to be a Boeing 767-200. The terminal is to have parallel concourses perpendicular to the runway centerlines, with nose-in aircraft parking. The top of the concourse will be 28 ft high at the end. The runways are 10,000 ft long.

- (a) Determine the separation requirements between runway centerlines, between the runway centerline and a parallel taxiway centerline, between parallel taxiway centerlines, between the runway and the terminal building line, between the runway and the property line, and between concourse faces.
- (b) Determine the area of the apron surface, in acres, between the concourses if six aircraft are parked on each terminal face.
- (c) Determine the property area, in acres, between the property line and the terminal building line from ends of the runways.
- (d) Determine the distance between the runway centerline and the aircraft holding line for takeoffs.

0.2 Answer

The design aircraft is the Boeing 767-200, which belongs to Aircraft Design Group V (ADG-V). FAA IFR separation standards for ADG-V aircraft are used.

(a) Separation Requirements

- Runway centerline to runway centerline:

$$\boxed{5000 \text{ ft}}$$

- Runway centerline to parallel taxiway centerline:

$$\boxed{400 \text{ ft}}$$

- Parallel taxiway centerline separation:

$$\boxed{267 \text{ ft}}$$

- Runway centerline to terminal building line:

$$400 + 267 + 160 + 190 = 1017 \approx \boxed{1020 \text{ ft}}$$

- Runway centerline to property line:

$$400 + 267 + 300 = 967 \approx \boxed{1000 \text{ ft}}$$

- Concourse face separation:

$$190 + 250 + 190 = \boxed{630 \text{ ft}}$$

(b) Apron Area Between Concourses Assuming a gate spacing of 170 ft per aircraft:

$$L = 6 \times 170 = 1020 \text{ ft}$$

Apron width:

$$W = 630 \text{ ft}$$

Apron area:

$$A = 1020 \times 630 = 642600 \text{ ft}^2$$

Converting to acres:

$$A = \frac{642600}{43560} \approx \boxed{14.8 \text{ acres}}$$

(c) Property Area Total width between terminal building line and property line:

$$1020 + 1000 = 2020 \text{ ft}$$

Runway length:

$$10000 \text{ ft}$$

Property area:

$$A = 2020 \times 10000 = 20.2 \times 10^6 \text{ ft}^2$$

Converting to acres:

$$A = \frac{20.2 \times 10^6}{43560} \approx \boxed{464 \text{ acres}}$$

(d) Runway Centerline to Holding Position For ADG-V precision IFR operations, the holding position distance is:

$$\boxed{280 \text{ ft}}$$